CASE STUDY ON WALMART U.S SALES ANALYSIS (2011 - 2014)

INTRODUCTION

This is a case study on Walmart, which is one of the largest retailers in the world. This is my capstone project, which involves using tools such as Microsoft Excel, SQL, Python, and Power BI. This dataset is a sample model of Walmart's operations in selected cities/states between 2011 and 2014.

There are six phases of analysis: Ask, Prepare, Process, Analyze, Share, and Act.

ASK PHASE

This Case study is on Walmart sales analysis which is an Public domain dataset available on Kaggle and GitHub repositories

Guiding questions

- What is the problem you are trying to solve?
- How can your insights drive business decisions?
- How can we identify trends and patterns in sales data across different stores and time periods?
- What factors contribute to higher sales performance in certain locations?

PREPARE PHASE

DATA ASSESSMENT ON THE PUBLIC DATASET

The Data is an public dataset which is stored in the website known as kaggle where we can compete ,collaborate and learn

THE ROCCC ASSESSMENT(Reliability,originality,comprehensive,citied and current)

Reliability :

This is an public dataset on Kaggle, where the data is available from 2011–2014 and it cannot be reliably used to analyze Walmart's sales and profit.

ORIGINALITY:

The Dataset is generated by the respondents via third party source.

COMPREHENSIVE

The dataset would be helpful in analyzing the sales, marketing strategy and profit of Walmart from 2011–2014

CITIED

it is mentioned in the kaggle that the dataset was made available by an third party source

CURRENT

The dataset was updated 2 years ago and it was downloaded by 620 people over the last 30 days

https://www.kaggle.com/datasets/gayatriwagadre/walmart-sales-analysis?resource=download

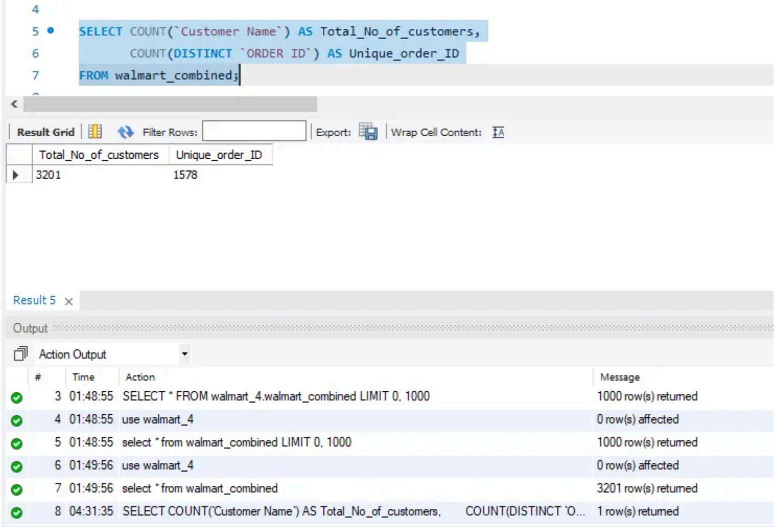
PROCESS PHASE

After looking at the Walmart 2011–2014 Excel data, I decided to remove blanks and duplicates if there were any. I explored the dataset, filtered and sorted where necessary, and after using Microsoft Excel for cleaning, I switched to MySQL for the remaining analysis. I changed the file format to CSV file (comma-delimited) and saved the file.

Next I opened MYSQL workbench and created a new schema or new database in the name of walmart_4 and then imported the table from excel and started analyzing the data.

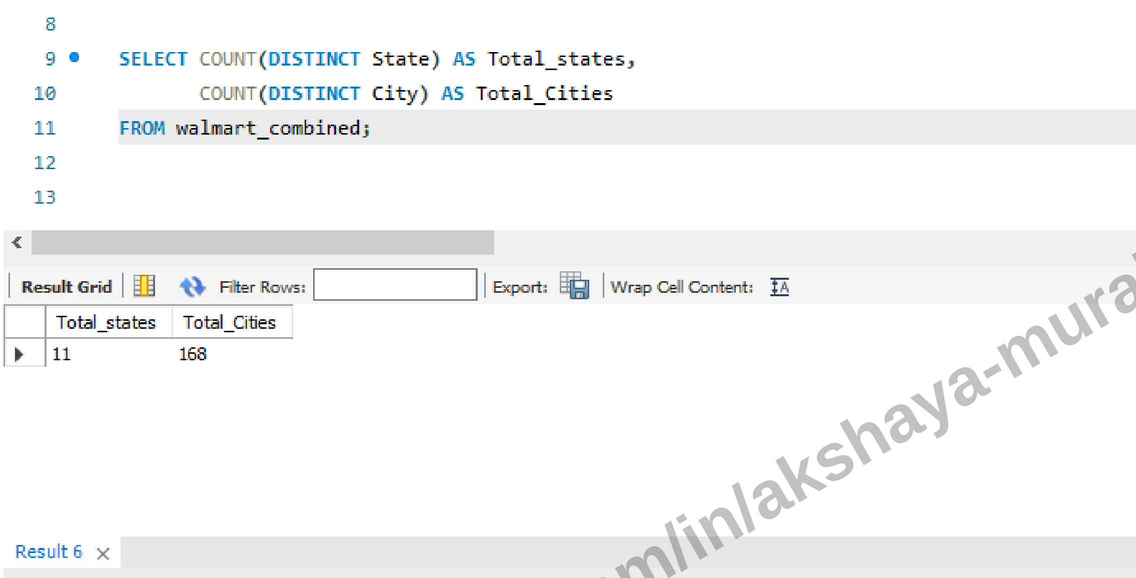
ANALYZE PHASE

First i began analyzing the Total Number of Customers and Total Number of Unique Order ID's S which gives me the clarity to understand the geographic spread of Walmart operations.



The total number of customers is 3,201 members from 2011–2014, and the distinct Order IDs received during this period total 1,578 orders.

Next i began checking count of unique state and city



As per the result, I am analyzing data from 11 total states in the United States. As we all know, Walmart is spread across more than 50 cities in the United States alone, and it generates revenue in millions and billions approximately. This is a sample dataset in which coding and results in MySQL Workbench show the geographic spread of Walmart's data, which is present in urban, semi-urban, and rural areas.

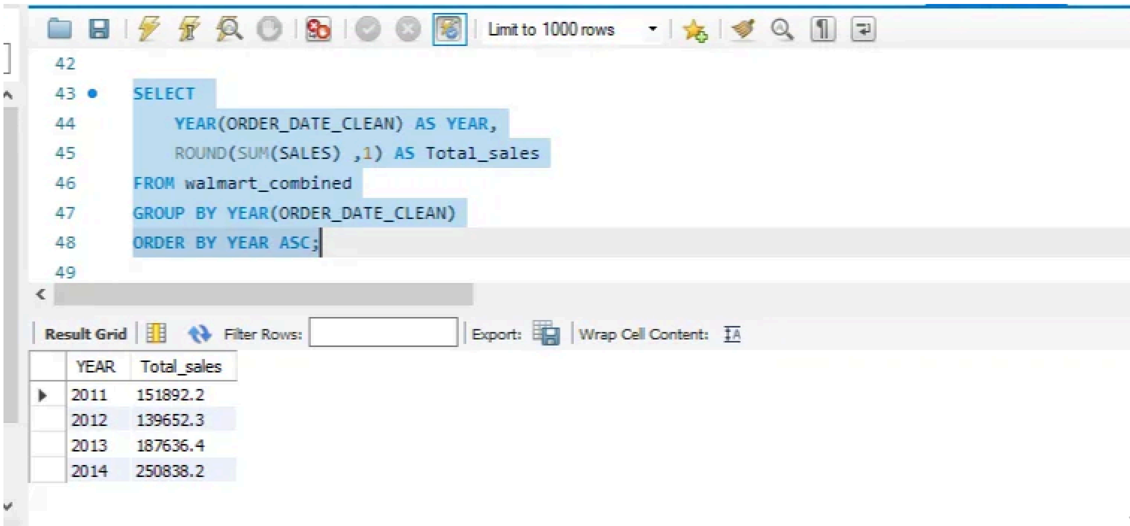
Next, I am going to analyze the most important part of analysis, which is sales, profit, and cost trends.

Total sales for this dataset is estimated to be $725,400 (2011-2014)

Total Profit for this dataset is estimated to be $108,400 (2011-2014)

Total Cost for this dataset is estimated to be $617,400 (2011-2014)

Now, I am breaking down sales and profit into four equal years to check if there is an increasing or decreasing trend.



so the Total sales for the year 2011 in this dataset is **$1,51,892.2**

in 2012 the Total sales goes on decreasing trend which is **$1,39,652**

in 2013 there is an increasing trend in sales which is **$1,87,636.4**

in 2014 also the sales goes on an huge increasing trend which is **$2,50,838.2**

Next i began analyzing profit for four equal years and Total profit %

(Profit = Revenue-cost)

As per my dataset sales is the revenue so it is (sales-cost)

so the Total Profit for the year 2011 in this dataset is **$20,065.7**

in 2012 the Total profit goes on a an consistent trend which is **$20,492**

in 2013 there is an increasing trend in profit which is **$23,959**

in 2014 the profit goes on an Huge increasing trend which contributes to **$43,900**

calculating Total profit percentage using the formula:

Total profit% = (Total sales/Total profit)*100

which gives an profit percentage of 14.9%

It represents an steady and scalable strategy across multiple states and cities

Next, I will analyze key summary statistics such as minimum, maximum, and average sales across different cities and states. This analysis will be performed using Python as part of the Share phase.

SHARE PHASE

As part of share phase i am using jupyter notebook to perform data analysis using python coding and in the first few screenshots i will be showcase insights about the walmart data using numpy,pandas and matplotlib through series of visualizations.

STEP 1: df.head()

```
[3]: import pandas as pd
     df = pd.read_csv(r"D:\walmart_combined.csv")
     df.head()
```

[3]:

	Order ID	ORDER_DATE	Ship Date	Customer Name	Country	City	State	Category	Product Name	Sales	Quantity	Profit	profit percentage	Cost per product	SHIPPING DURATION(DAYS)	order
0	CA-2013-138688	13-Jun-13	17-Jun-13	Darrin Van Huff	United States	Los Angeles	California	Labels	Self-Adhesive Address Labels for Typewriters b...	14.620	2	6.8714	14.9%	$8	4	
1	CA-2013-121755	16-Jan-13	20-Jan-13	Eric Hoffmann	United States	Los Angeles	California	Binders	Wilson Jones Active Use Binders	11.648	2	4.2224	14.9%	$7	4	
2	CA-2013-101343	18-Jul-13	23-Jul-13	Ruben Ausman	United States	Los Angeles	California	Storage	Eldon Base for stackable storage shelf, platinum	77.880	2	3.8940	14.9%	$74	5	
3	CA-2012-135545	24-Nov-12	30-Nov-12	Kunst Miller	United States	Los Angeles	California	Accessories	Verbatim 25 GB 6x Blu-ray Single Layer Recorda...	13.980	2	6.1512	14.9%	$8	6	

The dataset is previewed here using df.head() to know, understand and identify the key columns

STEP 2: df.info()

```
[17]: import pandas as pd
      df = pd.read_csv(r"D:\walmart_combined.csv")
      df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3201 entries, 0 to 3200
Data columns (total 16 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Order ID                3201 non-null   object 
 1   ORDER_DATE              3201 non-null   object 
 2   Ship Date               3201 non-null   object 
 3   Customer Name           3201 non-null   object 
 4   Country                 3201 non-null   object 
 5   City                    3201 non-null   object 
 6   State                   3201 non-null   object 
 7   Category                3201 non-null   object 
 8   Product Name            3201 non-null   object 
 9   Sales                   3201 non-null   float64
 10  Quantity                3201 non-null   int64  
 11  Profit                  3201 non-null   float64
 12  profit percentage       3201 non-null   object 
 13  Cost per product        3201 non-null   object 
 14  SHIPPING DURATION(DAYS) 3201 non-null   int64  
 15  order_date_clean        3201 non-null   object 
dtypes: float64(2), int64(2), object(12)
memory usage: 400.3+ KB
```

Using df.info() to preview the names and data types of each columns for cleaning and handling missing values
The dataset has 3201 records and 16 columns. Based on the
df.info() output, there are no missing values, and most columns are stored as object types. The numeric columns like Sales, Quantity, and Profit are correctly recognized as float64.

STEP 3: df.describe()

```
[19]: df.describe()
```

[19]:

	Sales	Quantity	Profit	SHIPPING DURATION(DAYS)
count	3201.000000	3201.000000	3201.000000	3201.00000
mean	228.059731	3.795376	34.205515	3.68010
std	528.094680	2.232919	175.113716	2.27846
min	0.990000	1.000000	-3399.980000	-7.00000
25%	19.440000	2.000000	3.895000	2.00000
50%	60.120000	3.000000	11.151000	4.00000
75%	215.976000	5.000000	33.158400	5.00000
max	13999.960000	14.000000	6719.980800	7.00000

using df.describe() to get the statistical summary of the data which are known as the highlighting metrics and is useful for predicting general trends for upcoming years.

Step 4 : groupby and sort_values() of states

```
[23]: df.groupby("State")["Sales"].agg(['min', 'max', 'sum', 'mean']).sort_values(by="sum", ascending=False)
```

[23]:

State	min	max	sum	mean
California	0.990	8187.650	450567.5915	232.012148
Washington	1.344	13999.960	153138.0280	259.996652
Arizona	1.408	1879.960	34284.9050	158.726412
Colorado	1.080	2549.985	31285.2360	175.759753
Oregon	1.080	1487.040	17282.1020	144.017517
Nevada	3.640	4535.976	16642.9020	437.971105
Utah	5.040	1499.950	10575.0360	225.000766
Montana	8.288	2999.950	5583.2560	398.804000
New Mexico	4.170	883.840	4783.5220	129.284378
Idaho	3.304	1128.390	4273.4860	224.920316
Wyoming	1603.136	1603.136	1603.1360	1603.136000

Here the Average ,Total,minimum,maximum and average sales is shown as per the states of USA which helps us showing current trends and forecast or predict future trends

Step 4 : groupby and sort_values() of Cities

```
[25]: df.groupby("City")["Sales"].agg(['min', 'max', 'sum', 'mean']).sort_values(by="sum", ascending=False)
```

[25]:

City	min	max	sum	mean
Los Angeles	2.880	4158.912	173168.865	236.893112
Seattle	1.344	13999.960	134320.440	261.323813
San Francisco	0.990	8187.650	110917.036	224.528413
San Diego	3.080	4476.800	47115.061	283.825669
Denver	1.872	1983.968	11690.089	271.862535
...	...	...	...	...
Lewiston	9.584	9.584	9.584	9.584000
Billings	8.288	8.288	8.288	8.288000
Auburn	4.180	4.180	4.180	4.180000
Everett	3.856	3.856	3.856	3.856000
San Luis Obispo	3.620	3.620	3.620	3.620000

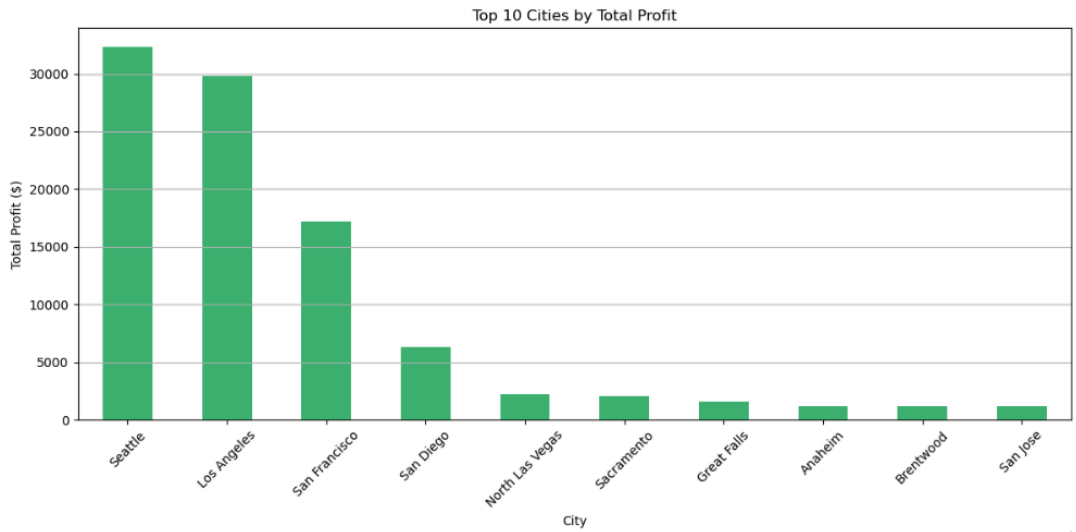
168 rows × 4 columns

"Among all 168 cities,
Los Angeles has the highest total sales (over $173K), followed by **Seattle** and **San Francisco**, highlighting the concentration of high sales in major urban areas."

Now through matplotlib i am filtering out the top 10 profitable cities and states with bar charts

```
[29]: import matplotlib.pyplot as plt

top_10_profit_cities = df.groupby("City")["Profit"].sum().sort_values(ascending=False).head(10)
plt.figure(figsize=(12,6))
top_10_profit_cities.plot(kind='bar', color='mediumseagreen')
plt.title("Top 10 Cities by Total Profit")
plt.ylabel("Total Profit ($)")
plt.xlabel("City")
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()
plt.show()
```



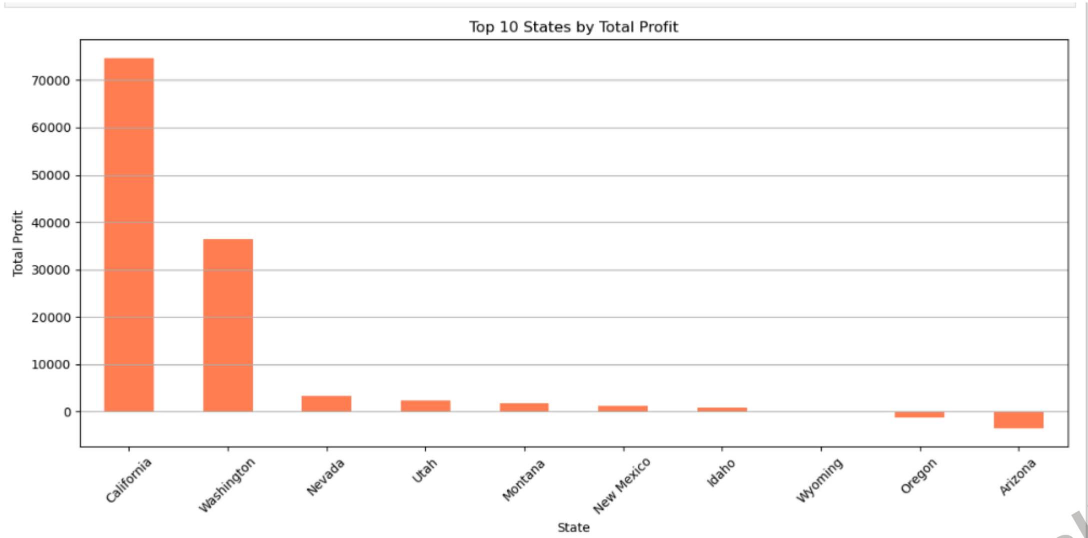
The most profitable city from 2011 to 2014 is **Seattle**, generating approximately **$30,000 in total profit**, followed by **Los Angeles**

.Based on this insight, I recommend **gradually expanding Walmart's operations in Seattle**, starting on a smaller scale. Given its strong historical performance, this strategic move could help increase Walmart's market presence and profitability in the region

Next i am filtering out the top 10 profitable cities through matplotlib with bar chart

[47]:
```
import matplotlib.pyplot as plt

top_10_profit_States = df.groupby("State")["Profit"].sum().sort_values(ascending=False).head(10)
plt.figure(figsize=(12,6))
top_10_profit_States.plot(kind='bar', color='coral')
plt.title("Top 10 States by Total Profit")
plt.ylabel("Total Profit")
plt.xlabel("State")
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()
plt.show()
```



so California is the Highest profitable state as per this data shown from 2011-2014

Next i am going to analyze this dataset with the final phase known as the ACT phase through series of POWER BI visualizations and forecast analysis for 2015 and 2016

ACT PHASE

First of all imported an table from excel to power BI and re-named it walmart_present

Using **Power Query**, I handled:

- Missing values
- Any potential data errors
- Finalized and loaded the cleaned dataset to the Power BI data model.

Second i created DAX measures for these columns:

Sales,

Quantity,

Profit

The DAX formulas are as follows :

Total Sales = sum(walmart_present[sales])

Total quantity = sum(walmart_present[quantity])

Total profit = sum(walmart_present[profit])

Total Cost = sum(walmart_present[cost_per_product])

Business Insights from the Dashboard

- 💰 **Total Profit:** $108.4K
- 🛒 **Total Sales:** $725.5K
- 📦 **Total Cost:** $617K

Next I moved to table view and created an new table through table tools (new table),so i copy pasted the order date table in the new table seperated the order date,month,year,quarter into columns .

The formulas are as follows :

ORDER_DATE_TABLE = calendar(FIRSTDATE(walmart_present[order_date]),LASTDATE(walmart_present[shipped_date]))

The order_Date_Table and an date column is created along with year and month

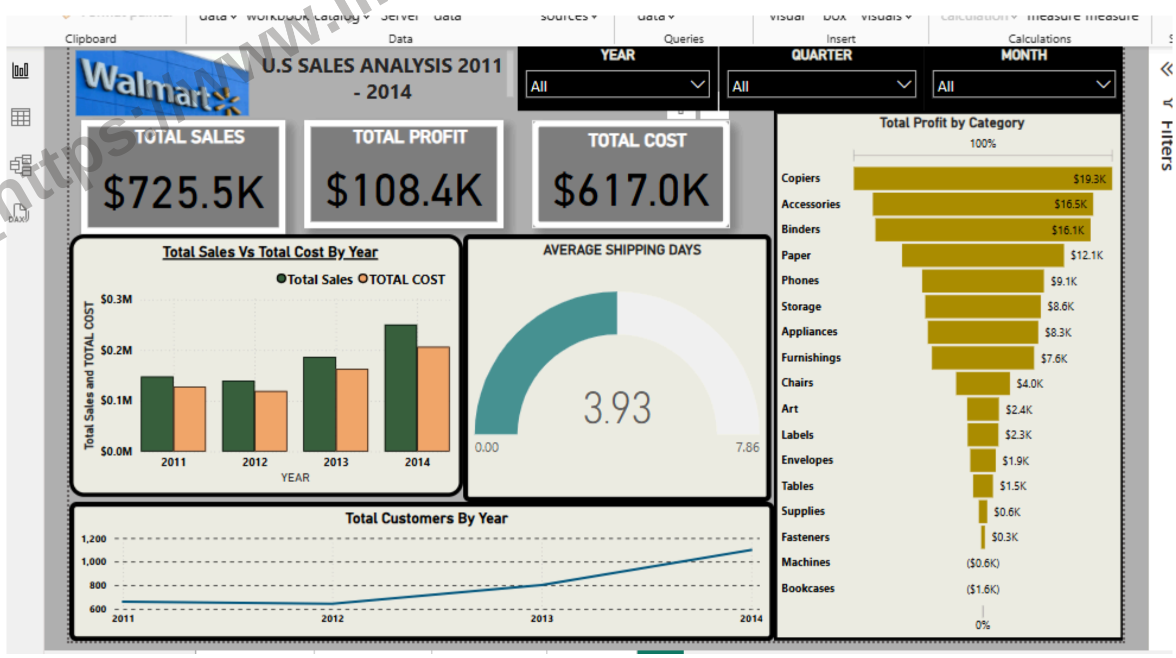
SEPERATE_YEAR = Year(ORDER_DATE_TABLE[Date])

SEPERATE_MONTH = Month(ORDER_DATE_TABLE[Month])

QUARTER = Quarter(ORDER_DATE_TABLE[quarter])

Next the most important DAX measure that would help the visual is the calculation of shipping duration in days which is:

SHIPPING_DURATION_DAYS = DATEDIFF(walmart_present[order_date],walmart_present[ship_date], DAY)



so the summary page(page 1) of my power BI report are as follows :

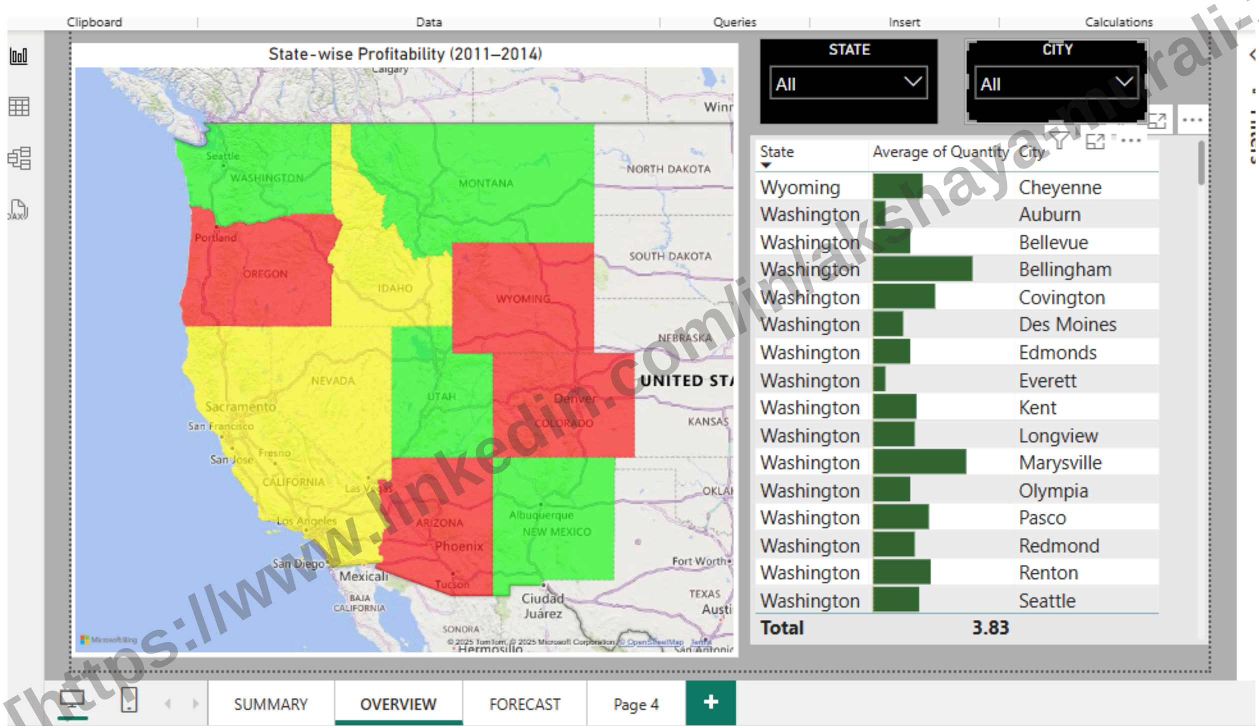
📍 **Top City**: Seattle – the highest profit contributor (~$30K)

📉**Loss-Making Categories**: Machines and Bookcases

📦**Top Categories by Profit**: Copiers, Accessories, Binders

⏱️ **Avg. Shipping Days:** 3.9

THE OVERVIEW PAGE

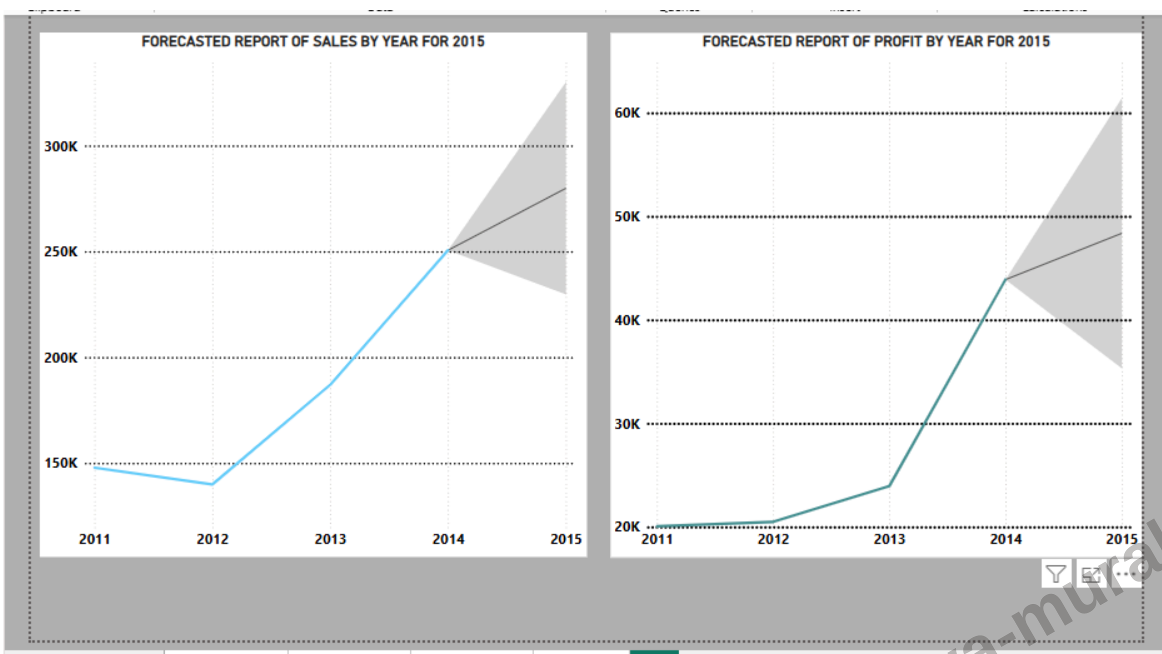


Created an column called as Profit Ratio and used an filled map visual to make an mark of state wise profitablity map

🗺️ **Geographic Insights**

- **Green States** performed well, while **Yellow States** flagged Average profits and **Red states** flagged low/No profits in the map

THE FORECAST PAGE



📈 Forecast for 2015 (via Line Charts)

- **Sales** and **Profit** both show a strong upward trajectory.
- With consistent growth, 2015 could see ~$300K+ in sales and ~$60K in profit.

🎯 Recommendation

Walmart should focus on Re-investing on smaller scale on profitable cities and improve marketing and product strategies in loss generating segments/categories